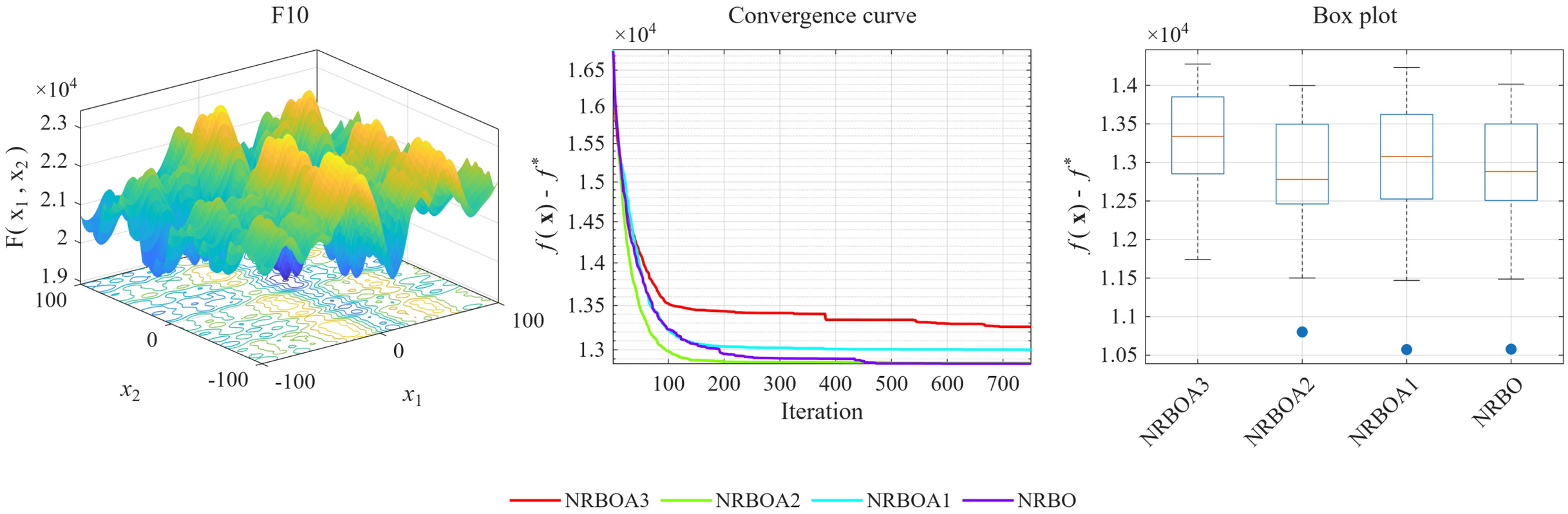

F10
×10^4
2.3
2.2
2.1
2
1.9
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
×10^4
1.65
1.6
1.55
1.5
1.45
1.4
1.35
1.3
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^4
1.4
1.35
1.3
1.25
1.2
1.15
1.1
1.05
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO